



Evaluation of X-ray computed tomography for quantifying macroporosity of loamy pasture soils



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ABSTRACT

Macropores are important pathways for rapid infiltration of water into soil as large soil pores allow roots, air, and water to penetrate into the soil. Soil compaction due to dairy cattle trampling may reduce soil macroporosity to the extent that growth or even survival of roots is limited by oxygen availability and soil strength. This study aims to evaluate the feasibility of X-ray computed tomography (CT) for determining macroporosity in dairy pasture soils and to assess the effects of sample volume (both within and between cores of varying sizes) and locations (i.e. site and soil depth) on measures of porosity. Undisturbed soil cores (50 and 65 mm diameter) were collected from two depths (0 to 10 cm and 20 to 30 cm) and from three sampling sites (representing moderately to highly trafficked zones) within a commercial dairy paddock. The intact soil cores were scanned at three resolutions (30, 109 and 138 μm) using X-ray CT, and porosity and mean pore diameter measured in various volumes of these scans. Porosity and mean pore diameter decreased with depth but did not differ across the three sampling sites; this was consistent with measures of porosity based on a traditional bulk density based method and soil water retention curves. There was no significant ($P > 0.05$) effect of analysing porosity at increasing distances from the core edge. Likewise, increasing the volume of soil that was analysed within each core was not found to have a significant ($P > 0.05$) effect on macroporosity. However, mean pore diameter was found to significantly ($P < 0.05$) increase with increasing volume of soil measured (both within a soil core and with increasing core size) and significantly ($P < 0.05$) decrease with increasing resolution of the scans. The results suggest that while absolute measures of macroporosity might not change with core size or the volume of soil analysed, the pore-space characteristics that are captured differ significantly. Macroporosity values for various pore size classes (0.2 to 298 μm pore diameters) assessed using soil–water retention curves compared with those determined using the X-ray CT were found to be comparable. Consequently, X-ray CT is a valuable tool for characterising pore-space from the macro- to the micro-scale, however, sampling and analysis strategies must be appropriate for the specific research aims. The practical implications of the results are discussed.

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1. Introduction

The influence of macroporosity i.e. pores larger than 30 μm in diameter, on soil infiltration, water retention and movement of agrochemicals and bacteria through soil profiles and subsequently into the sub-surface is well recognised around the world (Allaire-Leung et al., 2000; Amer et al., 2009; Anderson et al., 2010; Asare et al., 2001; Perret et al., 1999). Understanding the influence of soil macropore distribution on aeration and soil water relationships is important for evaluating root growth and root respiration. In grazing systems, maintaining pasture growth is integral to productivity. However, soil compaction due to livestock grazing can be substantial (Betteridge et al., 1999; Chaichi et al., 2005; Drewry et al., 2008; Greenwood and McKenzie, 2001; Greenwood et al., 1997; Singleton et al., 2000; Stavi et al., 2011), and

this reduces soil macroporosity (e.g. Krümmelbein et al., 2009) and breaks vertical pore continuity (Greenwood and McKenzie, 2001; Reszkowska et al., 2011) through the disruption of aggregates into smaller particles and the repacking of smaller particles to fill existing voids (Cattle and Southorn, 2010). Soil compaction may reduce macroporosity to the extent that growth or even survival of roots is limited by oxygen availability and high soil strength.

Traditionally, the soil–water retention method has been used widely around the world for characterising the macroporosity and pore-size distribution of soils (e.g. Amer et al., 2009; Rachman et al., 2005; Stingaciu et al., 2010; Vogel, 2000). This method involves laboratory measurement of soil–water content at various soil–water matric potentials and inferring porosity using the relationship between pore diameter and liquid flow. However, this method does not provide distribution of pores larger than 300 μm in diameter nor their continuity.

Advances in the application of computed tomography (CT) for evaluation of soil porosity have added a new research tool for investigating

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macroporosity distribution and pore diameters (Anderson et al., 1990, 2010; Borges et al., 2012; Cortina-Januchs et al., 2011; Kim et al., 2010; Munkholm et al., 2012, 2013; Taina et al., 2008; Vaz et al., 2011). Computed tomography is a non-invasive imaging technique that allows high resolution, three-dimensional, non-destructive imaging of heterogeneous soils and permits actual, rather than inferred, characteristics of soil pores (Cortina-Januchs et al., 2011; Grevers et al., 1989). X-ray CT scanning methods have been used effectively for measuring pore size, shape, distribution and arrangement of soil pores, surface area and pore connectivity (Kumar et al., 2010; Udawatta et al., 2008). CT methods provide a fine resolution of measurement, on a millimetre to micrometre-scale (Gantzer and Anderson, 2002; Kim et al., 2010; Kumar et al., 2010) and hence have the potential to detect small differences in pore geometry and other spatial parameters.

Despite the value of CT as a technique for understanding the impacts of animal compaction on soil macroporosity and subsequent implications for pasture productivity, very limited information is available (Pires et al., 2004) on the optimum sampling strategies (both within and between intact soil cores) for measuring pore characteristics of pasture soils using X-ray CT. There is also a lack of information available (but see Rachman et al., 2005; Stingaciu et al., 2010; Vogel, 2000) on the comparison of CT-measured and soil–water retention method soil porosity. The objectives of this study were to: (i) evaluate the effects of sample core diameters and spatial resolutions on porosity characteristics of loamy pasture soils using X-ray CT technology; (ii) quantify the effect of sample volume and edge effect on porosity characteristics using CT technology; (iii) determine vertical variations in porosity characteristics within collected samples and compare with whole-sample-measured values; and (iv) compare porosity measured by X-ray CT technology with porosity measured by traditional methods.

2. Materials and methods

2.1. Site description

This study, which took place in spring 2011, was conducted on a 2 ha paddock of a commercial grazing system dairy farm located 150 km south east of Melbourne, south eastern Australia (38° 13' S and 145° 48' E). The paddock elevation ranged from 114 to 126 m above sea level and the landform comprised undulating rises. The soil type of the paddock, classified according to Australian Soil Classification System (Isbell, 2002), is a Brown Dermosol and according to Word Reference Base System (IUSS Working Group WRB, 2007) is Umbric Alisol. This soil is formed mostly on cretaceous parent material with some tertiary and paleozoic sediments. The long-term mean annual rainfall, for the period 1902 to 2011, recorded 10 km away from the study site was 1004 mm. The mean annual temperature is 18 °C and can range from 0 °C in winter to 40 °C in summer.

In the grazing systems typical of this region, the herd spends the majority of their time grazing pasture paddocks and typically about 10% of their time in the milking facility. The study paddock has been used as one of the 'night' paddocks, for 15 years i.e. where the milking herds are held between the evening and morning. In 2007 the paddock was resown to a permanent pasture after first ploughing and sowing a summer crop of turnips (*Brassica* spp) in the previous spring (2006). The pasture was dominated by perennial ryegrass (*Lolium perenne* L.) with low white clover (*Trifolium repens* L.) content. Lime was applied at 1 tonne ha⁻¹ when the turnip crop was planted. Urea has been applied annually at a rate of 70 kg ha⁻¹ while phosphorus and potash were applied as 1 & 1 fertiliser (4.4% P, 25.0% K, 5.5% S and 9.5% Ca) at 250 kg ha⁻¹ annually from 2001 to 2009. Silage has been harvested from this paddock every year for the past ten years.

2.2. Soil sampling

Three sites were selected within the study paddock: gateway, upper slope, and lower slope. At each site a 5 m × 5 m area was established and within this area eight sampling locations were selected randomly by throwing a pencil backward. Soil samples were collected using 50 mm PVC tubes (50 mm diameter × 63 mm long, with a wall thickness of 3 mm) at four of the eight locations and at the other four locations soil samples were collected using 65 mm PVC tubes (65 mm diameter × 63 mm long × 3 mm). At each sampling location, the intact soil samples were collected from the 0 to 10 cm and 20 to 30 cm soil depths giving a total of 48 samples. The PVC tubes were driven into the soil gently with a falling weight hand corer. The intact core samples were trimmed, sealed in plastic and stored at room temperature (22 °C) until scanned. Macroporosity and pore diameters were measured in the 48 intact cores using CT. At each of the sampling locations, additional intact samples were taken at corresponding soil depths using 73 mm diameter by 63 mm long brass cores for measuring porosity characteristics using a soil water retention method.

Additional soil was collected at each core location for analysis of routinely measured soil properties. Mean values of some physical and chemical properties for three sampling sites at two soil depths are presented in Table 1. Clay content did not vary between the three sites (gateway, upper slope or lower slope) or the two soil depths; silt content increased and coarse sand content decreased with depth only at the gateway. Fine sand was the dominant soil fraction (54–59%), followed by silt content (25–29%). Soil texture, determined according to the ISSS system (Leeper, 1974), showed that silty loam was the dominant soil texture at both soil depths. Most chemical properties did not vary between sites but varied with depth. Total carbon (TC) content was consistently higher in the 0 to 10 cm than in the 20 to 30 cm depth. Calcium and Na were higher in the upper soil depth but EC, pH and K were similar at both depths. Calcium was the major cation in these soils.

2.3. Porosity characteristics using X-ray CT

2.3.1. X-ray CT system and image analysis procedures

Intact soil cores were scanned using a Vtomex system (GE Phoenix, Germany) fitted with an X-ray microfocus tube (240 kV source, 4 µm spot size, tungsten reflective target) and a 512 × 512 pixel array detector. Cores were scanned using a full 360° rotation of the sample. Digital

Table 1

Mean soil chemical and physical properties at three sites (gateway, upper slope, lower slope) and two soil depths in the study paddock.

	Site					
	Gateway		Lower slope		Upper slope	
	Depth (cm)		Depth (cm)		Depth (cm)	
	0–10	20–30	0–10	20–30	0–10	20–30
<i>Soil property</i>						
Clay (%)	12	12	11	10	11	11
Silt (%)	25	29	25	26	25	25
Fine sand (%)	55	54	59	58	58	59
Coarse sand (%)	8	4	5	6	5	5
Total sand (%)	63	59	64	64	64	64
TC (%)	5.1	1.4	4.4	1.0	4.3	2.0
EC (dS/m)	0.1	0.1	0.2	0.1	0.1	0.1
pH (water)	5.2	4.8	4.8	5	5.4	4.9
pH (CaCl ₂)	4.6	4.2	4.4	4.4	4.7	4.3
<i>Exchange cations (cmol⁽⁺⁾ kg⁻¹)</i>						
Ca	3.6	1.0	3.4	0.93	3.6	1.1
Mg	1.2	0.4	0.8	0.3	1.1	0.4
K	0.5	0.1	0.2	0.1	0.2	0.2
Na	0.2	0.1	0.1	0.1	0.1	0.1
Sum of four cations	5.4	1.6	4.6	1.3	5	1.7

TC, total carbon.

radiographs were reconstructed using phoenix datosx reconstruction software. For imaging a subset of the “full core” scans, a “region of interest” reconstruction was selected in the reconstruction software (v1.5.0.22-64 bit; 2008 GE Sensing and Inspection technologies GmbH). Tomographs were imported into VG Studio MAX v 2.0 (Volume Graphics, Germany) and exported as a stack of voxel-thick TIFF images for export into the processing software Fiji (a distribution of ImageJ; Rasband, 2002; Schindelin, 2008). Stacks were initially cropped to remove slices at the top and bottom of the core that were subject to field-of-view artefacts from the cone beam source geometry. Stacks were then further cropped in a square around the centre of the horizontal plane for porosity analysis (Table 2). The cross-sectional area of these sub-samples in the horizontal plane was determined by the maximum possible dimensions for a square that could fit within the corresponding core diameters. This region excluded voids near the core walls and minimised any effects of beam hardening around the edge of the scanned region. The stack histogram was normalised using the Enhance Contrast function. Voxels were classed as pore-space or soil medium using the ‘Default’ auto-threshold (a variant of Iso-data; Ridler and Calvard, 1978). A visual comparison of the macro-porosity in the thresholded image with the macro-porosity in the original grey-scale image was made at 3 slices in the stack (upper, middle and lower) and the greyscale values manually adjusted on a sample by sample basis to reduce any over- or under-estimation of macro-pore space. Good contrast between solid and pore-space was achieved in both “full core” and “region of interest” scans and thresholding was adjusted to enable a best fit of macro-pore space. As with all methods of thresholding processes, this involved a degree of user subjectivity (Baveye et al., 2010). Porosity and pore-size distribution were analysed using the plugin SCAMP (SIMBIOS Centre, University of Abertay, Dundee). Henceforth, CT-based measures of macroporosity (CT macroporosity) refer to the volume of pores with radii equal to or greater than that of the given scanning resolution.

2.3.2. Effect of core diameter, resolution, image volume and scan location on measures of porosity

The 50 and 65 mm cores were scanned at 30; 109; and 138 μm resolution and porosity measured in various volumes of these different scans to investigate (a) the relationship between resolution and core diameter and (b) how sample volume, scan location and scanning resolution within a core affect measures of macroporosity. The 73 mm diameter cores were not scanned as these were for the purpose of traditional soil water retention methods and X-ray energy was insufficient to penetrate the metal core casing.

2.3.2.1. Core diameter and resolution. Resolutions and scanning parameters were initially optimised for each of the two core diameters (Table 2). As resolution is linearly related to sample diameter, scanning resolutions of 109 and 138 μm for the 50 and 65 mm diameter

cores, respectively, were identified as being the highest resolution that could be achieved while imaging the full diameter of the given core. Cores were scanned at these respective resolutions. The full diameter of the 65 mm cores with length of 50 mm were also scanned at 109 μm for comparison of macroporosity with the 50 mm diameter cores at that resolution. The pores detected at each of these two resolutions were divided into five fractions according to their diameter (Glab, 2007): >2000 μm , 1000–2000 μm , 600–1000 μm , 300–600 μm and 218–300 μm (for 109 μm resolution) or 276–300 μm (for 138 μm resolution).

2.3.2.2. Image volume. Additional analyses were undertaken to investigate the effect of sample volume within a core and potential edge effects from soil coring on porosity. For these analyses the 0 to 10 cm cores were selected as they had the highest porosity and hence the greatest likelihood of detecting sample volume and “edge” effects on porosity. As “edge” effects were being investigated, the scans for the full diameter of the cores were used (i.e. 109 μm scans for the 50 mm cores; 138 μm scans for the 65 mm cores). To measure the effect of sample volume on porosity, the thresholded images from the original rectangular prism analysed for each core were cropped into a series of five concentric rectangular prisms with a standard vertical height of 400 slices (Fig. 1.). The five concentric rectangular prisms represented 100% (Volume 1); 80% (Volume 2); 60% (Volume 3); 40% (Volume 4) and 20% (Volume 5) of the total sample volume. The porosity measured for each of these volumes was then subtracted from that of the adjacent interior concentric rectangular prism to determine the effect of distance from the core edge on porosity (i.e. porosity of V1–V2; V2–V3; V3–V4; V4–V5 giving four equal volumes that represented 20% volume of the original).

2.3.2.3. Scan location. The 50 mm diameter cores were selected for additional higher resolution scans to (a) compare how scan resolution within a core affects measures of porosity and (b) compare CT-based measures of pore-size distribution with that determined using a traditional water retention method. A scan resolution of 30 μm was the

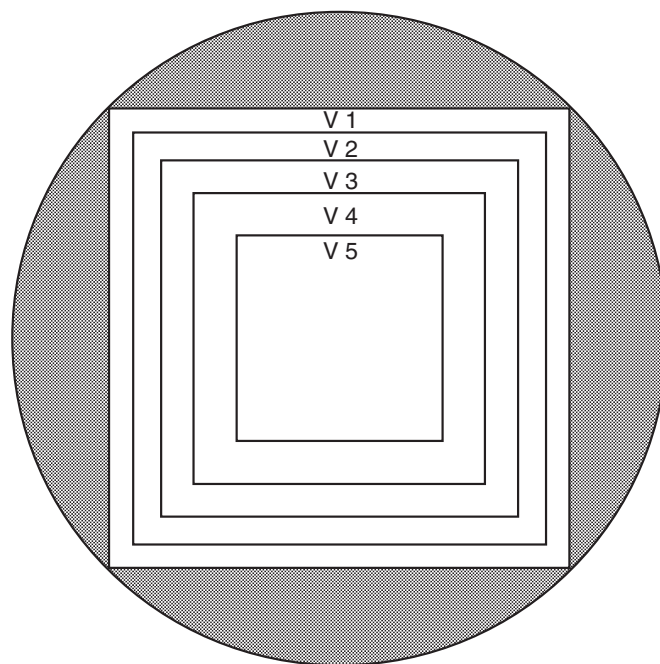


Fig. 1. Schematic diagram (not to scale) of the horizontal plane of a soil core demonstrating the concentric squares used for soil image analysis. The height of the sample in all five cases was 48 mm. V1, 100% of total image volume; V2, 80% of total image volume; V3, 60% of total image volume; V4, 40% of total image volume; and V5, 20% of total image volume. These smaller regions were selected after thresholding the original images.

Table 2
X-ray scanning and CT image analysis parameters.

Parameters	Core diameter (mm)			
	65		50	
	Isotropic voxel size (μm)		Isotropic voxel size (μm)	
	109	138	109	30
No of images per scan	1200	1200	1200	1200
Filter	Copper	Copper	Copper	Copper
Filter width (mm)	0.5	0.5	0.5	0.5
X-ray voltage (kV)	170	160	160	180
Current (μA)	160	150	150	150
Timing (ms)	200	200	200	400
Cropped dimensions of horizontal plane analysed (mm)	34.4 × 34.4	43.6 × 43.6	34.5 × 34.5	10.5 × 10.5

highest resolution that could be achieved and was used for a “region of interest” of 10.5×10.5 mm from the centre of the 50 mm cores. Three $30 \mu\text{m}$ scans were taken at three different vertical positions within each 50 mm diameter core; one at the centre of the core and two others equidistant (17 mm) above and below the centre of the core (Table 2). The resultant scanning depths for the three scanning positions were 14.5, 31.5 and 48.5 mm and 214.5, 231.5 and 248.5 mm respectively for samples collected from the 0 to 10 and 20 to 30 cm soil depths. The three scan locations were used to investigate how arbitrary sampling of porosity in the middle of a core may bias results.

2.4. Porosity characteristics using traditional methods

Pore size distributions were determined for additional intact soil samples collected in brass cores using water retention curves following Turner et al. (1984) and the relationship between pressure head and pore diameter:

$$h = \frac{0.298}{d} \quad (1)$$

where h is the capillary pressure head (cm); and d is the diameter of the pore size (cm). Using Eq. (1), the pore size diameters (298, 149, 60, 30, 10, 5, 3, 1 and $0.2 \mu\text{m}$) were calculated for h values of -10 , -20 , -50 , -100 , -300 , -600 , -1000 , -3000 and $-15,000$ matric potential

respectively. Soil-water content (SWC) of the undisturbed samples was determined at -10 , -20 , -50 , and -100 cm matric potential using ceramic plate bubble tower suction tables and at -300 , -600 , -1000 , and -3000 cm matric potential using a pressure plate apparatus. The intact core samples were saturated slowly by immersing the samples in a waterbath (to two-thirds of the length of the sample from the bottom) containing 0.01 M CaCl_2 to reduce dispersion of clay particles (Halliwell et al., 2001) and maintain soil structure (Kumar et al., 2010) and equilibrated to the given matric potential. After equilibrating at a matric potential of -3000 cm, the samples were oven-dried at 105°C for 48 h to determine bulk density (calculated as the oven dried mass of soil divided by the volume of the soil core) and volumetric soil-water contents at the above eight matric potentials. Soil-water content at $-15,000$ cm matric potential was estimated for the disturbed and sieved ($<2 \text{ mm}$ diameter) samples using a 15-bar ceramic plate. Total porosity (TPS; $\text{m}^3 \text{ m}^{-3}$) was calculated as $1 - \text{BD} / 2.65$, assuming a particle density of 2.65 Mg m^{-3} . Air-filled porosity at each of the above matric potentials was calculated as the difference between TPS and SWC. The pores larger than $30 \mu\text{m}$ diameter are considered as macropores in this study.

Porosity values were also estimated for CT samples. After CT-analysis, samples were oven-dried, TPS and volumetric soil water content at the time of sampling (SWC_f) were determined and air-filled porosity was calculated as the difference between TPS and SWC_f .

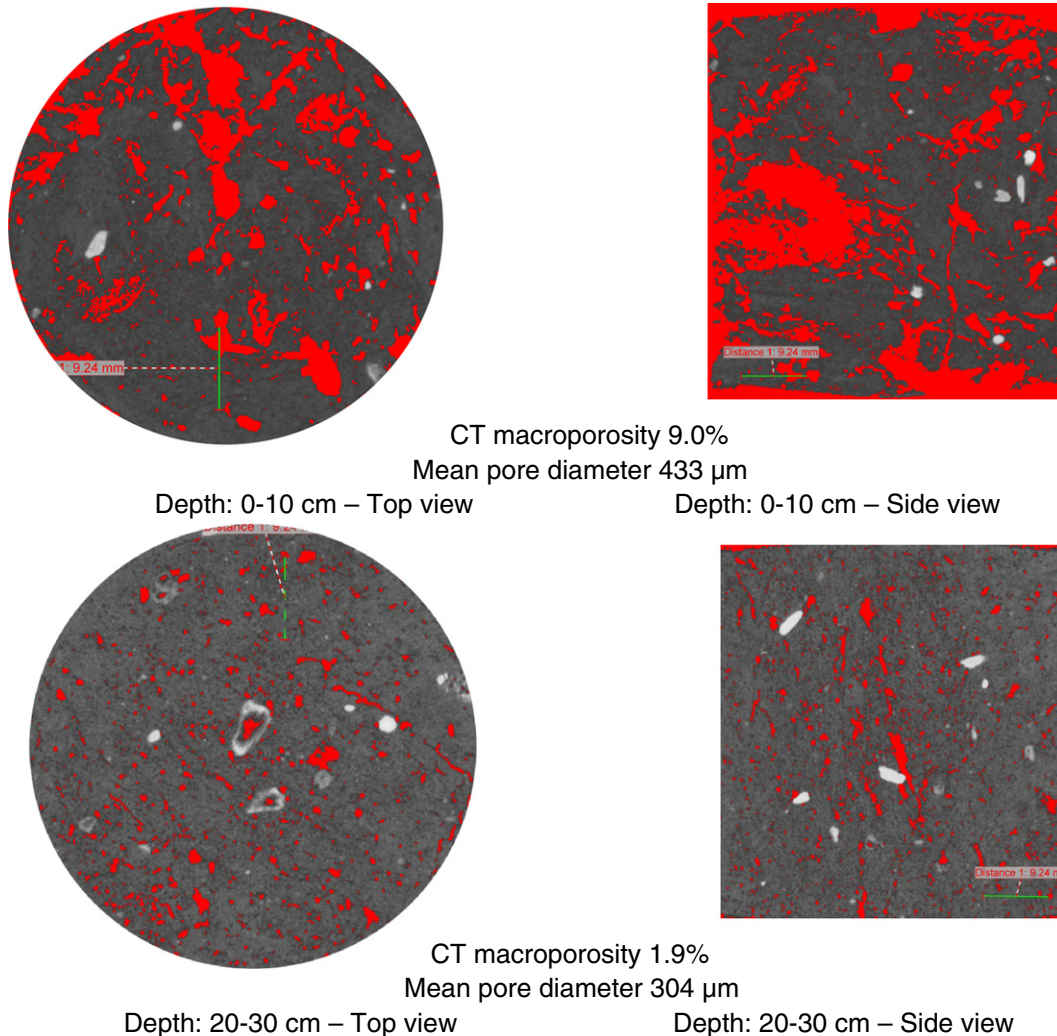


Fig. 2. Visual examples of top and side view of images from one sampling location at two soil depths with corresponding estimates of CT macroporosity and mean pore diameter.

2.5. Statistical analysis

2.5.1. Site by depth by core diameter and scanning resolution

Results for porosity, pore diameter, and porosity within in each of 5 pore diameter classes, were each analysed using a linear mixed model in REML (Patterson and Thompson, 1971) in GenStat 13th Edition (VSN International Ltd.). Since data were unbalanced, a factor, "DiamRes", was defined for the 3 combinations of core diameter and scanning resolution, namely, "core diameter 50 mm, resolution 109 μm ", "core diameter 65 mm, resolution 109 μm " and "core diameter 65 mm, resolution 138 μm ". Factorial fixed effects in the mixed model were then defined as Site $\times$ Depth $\times$ DiamRes. Random effects were specified as nested: sampling points (24) and depth (two levels) within sampling points, and measurement (two levels of resolution) within each depth within (some) sampling point. Analysis of deviance F-tests was used to test the main effects and interactions among the fixed effect factors. The main effect of core diameter was tested by defining a contrast between the predicted means for diameters 50 and 65 mm at resolution 109 μm . The main effect of resolution was tested by defining a contrast between the predicted means for resolutions 109 μm and 138 μm at core diameter 65 mm. Interactions for these effects with site and/or depth were tested using analogous contrasts (in this case, differences of differences) of predicted means. The porosity and pore diameter data were log-transformed to meet analysis assumptions of normal distribution and constant variance.

2.5.2. Image volume

Porosity and pore diameter volume data measured in the 0 to 10 cm depth were analysed separately for 50 mm and 65 mm cores. These were analysed by using a linear mixed model in REML. Fixed effects in the mixed model were defined as Site $\times$ Image volume and random effects were specified as nested: sampling points (12) and points within site.

2.5.3. Scan position

Porosity and pore diameter for three scanning positions within a core and for the whole core, measured using 50 mm cores at the 0 to 10 and 20 to 30 cm depths, were analysed separately for each of these two depths using a linear mixed model in REML. Fixed effects in the mixed model were defined as Site $\times$ Scan position and random effects were specified as nested: sampling points (12) and points within site.

3. Results

3.1. Porosity characteristics measured using the CT scan method

3.1.1. Effect of site and depth on porosity characteristics

Significant ($P < 0.001$) differences in macroporosity values measured using CT were observed between soil depths (Fig. 2, Table 3). Soil CT macroporosity and mean pore diameter were significantly greater in the 0 to 10 cm depth compared to the 20 to 30 cm depth. No significant ($P > 0.05$) differences in CT macroporosity or mean pore diameter were found across the three sampling sites.

3.1.2. Effect of soil core diameter and scanning resolution on porosity characteristics

For scanning resolutions that captured the full diameter of the soil cores (Table 3), the CT macroporosity measured at 109 μm was marginally different ($P = 0.048$) between two core diameters. On the other hand, CT macroporosity measured in the 65 mm cores at 138 μm was not significantly different to that of the 50 mm cores measured at 109 μm ($P > 0.05$). The 65 mm cores were subsequently scanned at 109 μm and the CT macroporosity measured was significantly greater than that measured for the 65 mm cores at 138 μm ($P < 0.001$) but not significantly greater than the 50 mm cores scanned at 109 μm ($P = 0.063$). In both instances, mean pore diameter was larger for the 138 μm scan than the 109 μm scans ($P < 0.001$).

At both soil depths CT macroporosity values for each of these three pore diameter classes <300, 300–600 and 600–1000 μm were significantly ($P < 0.001$) greater for scans at 109 μm compared with those at 138 μm (Table 4). No significant ($P > 0.05$) differences in CT macroporosity values were found for pore diameter classes of 1000–2000 and >2000 μm .

3.1.3. Effect of volume of image analysed and core edges on porosity characteristics

No significant ($P > 0.05$) differences were found in the CT macroporosity measured in volumes ranging from 20 to 100% of the original volume analysed for both the 50 and 65 mm diameter cores collected at 0 to 10 cm (Table 5). On the other hand, mean pore diameter decreased significantly ($P = 0.002$) as the volume of the soil that was analysed in the 50 mm diameter cores decreased. The effect of the core edge was assessed by subtracting CT macroporosity measured in adjacent interior volumes (see Fig. 1, Table 6). No significant ($P > 0.05$) differences were found in CT macroporosity between these

Table 3
Effect of soil sample core diameters (50, 65 mm) and CT scan resolutions (109, 138 μm) on porosity and mean pore diameter at three sampling sites (gateway, upper slope, lower slope) and two depths (0 to 10, 20 to 30 cm).

Effect of soil depth						Tests for fixed effects		
Soil property	Depth			SEd		Fixed term	F prob	
	0–10	20–30	SEd				CT macroporosity ($\text{m}^3 \text{m}^{-3}, \%$)	Pore diameter (μm)
ln(CT macroporosity, $\text{m}^3 \text{m}^{-3}, \%$)	2.42	1.66	0.096			Site	0.258	0.931
CT macroporosity ($\text{m}^3 \text{m}^{-3}, \%$)	11.2	5.3				Depth	<0.001	<0.001
ln(mean pore diameter, μm)	6.11	5.80	0.025			DiamRes	<0.001	<0.001
Mean pore diameter (μm)	448.1	328.7				Diam (@Res109)	0.048	0.778
						Res (@Diam 65)	<0.001	<0.001
Effect of soil sample core diameter and scanning resolutions						Site $\times$ depth	0.056	0.622
Sample diameter (mm)	50	65	65	SEd _D		Site $\times$ DiamRes	0.489	0.825
Resolution (μm)	109	109	138		SEd _R	Depth $\times$ DiamRes	0.083	0.656
						Diam $\times$ depth (109)	0.032	0.551
ln(CT macroporosity, $\text{m}^3 \text{m}^{-3}, \%$)	1.94	2.29	1.88	0.165	0.070	Resolution $\times$ depth (65)	0.742	0.296
CT macroporosity ($\text{m}^3 \text{m}^{-3}, \%$)	7.0	9.9	6.6			Site $\times$ depth $\times$ DiamRes	0.870	0.497
ln(mean pore diameter (μm))	5.89	5.87	6.09	0.0268	0.0215			
Mean pore diameter (μm)	361	354	441					

F prob, probability values for fixed terms are from Wald tests while for remaining model terms the probability values are from t-tests. SEd_D, standard error of difference for comparing diameter (Diam) at resolution (Res) 109; SEd_R, standard error of difference for comparing resolution at diameter 65.

Table 4

Effect of CT scan resolutions on pore size distribution of whole core samples at two soil depths.

Resolutions	Pore diameter class (μm)	Mean porosity ($\text{m}^3 \text{m}^{-3}$, %) at two depths	
		Depth (cm)	
		0–10	20–30
109- μm	>2000	0.5	0.1
	1000–2000	3.4	0.8
	600–1000	3.5	1.5
	300–600	3.0	1.7
	218–300	3.8	2.2
138- μm	>2000	0.9	0.1
	1000–2000	3.5	0.9
	600–1000	2.4	1.1
	300–600	1.8	1.2
	276–300	2.5	1.5

four equal volumes as the distance from the edge of the core to the region of soil analysed increased.

3.1.4. Effect of scanning position on porosity characteristics measured within soil cores

The CT macroporosity and mean pore diameter of higher resolution (30 μm) scans taken at the top, middle and bottom of each of the 50 mm cores are presented in Table 7. In the 0 to 10 cm soil cores, CT macroporosity and mean pore diameter were significantly lower ($P < 0.05$) in the bottom position than the top/middle scan positions. CT macroporosity measured in the whole core (at 109 μm) was equivalent to that measured in the top and middle scans but mean pore diameter was significantly higher for the whole-core scans. In the 20 to 30 cm soil depth, no significant differences were measured in CT macroporosity or mean pore diameter between the three scan locations. While CT macroporosity of the whole-core did not differ significantly ($P > 0.05$) to any of the three higher resolution scans, mean pore diameter in the whole-core analysis was significantly greater ($P < 0.001$).

The effect of resolution on pore size distribution is presented in Fig. 3. The range in pore radius was greater for cores scanned at

Table 5

Mean macroporosity and pore diameter for different volumes of image analysed for scans of soil cores of 50 and 65 mm diameter collected at the 0 to 10 cm soil depth.

Core diameter (mm)	Soil property	Volume	Mean	Fixed term	Av SED	F pr
50	CT macroporosity ($\text{m}^3 \text{m}^{-3}$, %)	V1	9.5	Site	4.3	0.607
		V2	9.4	Volume	0.3	0.823
		V3	9.3	Site $\times$ volume	3.2	0.298
		V4	9.3			
		V5	9.2			
	Mean pore diameter (μm)	V1	404	Site	20	0.403
		V2	400	Volume	3	0.002
		V3	398	Site $\times$ volume	17	0.503
		V4	395			
		V5	390			
65	CT macroporosity ($\text{m}^3 \text{m}^{-3}$, %)	V1	11.4	Site	3.4	0.045
		V2	11.5	Volume	0.4	0.183
		V3	11.8	Site $\times$ volume	2.7	0.779
		V4	12.0			
		V5	12.3			
	Mean pore diameter (μm)	V1	519	Site	43	0.637
		V2	517	Volume	4	0.696
		V3	515	Site $\times$ volume	33	0.964
		V4	514			
		V5	514			

Scan resolution of 109 and 138 μm for 50 and 65 mm cores respectively.

V1, 100% of total volume; V2, 80% of total volume; V3, 60% of total volume; V4, 40% of total volume; V5, 20% of total volume (Fig. 1).

F prob, probability values for fixed terms are from Wald tests.

Table 6

Effect of volume increments of soil image analysed on macroporosity for two diameters (50, 65 mm) of soil core collected at the 0 to 10 cm soil depth.

Core diameter (mm)	ΔV	CT macroporosity (m^3/m^3 , %)	Fixed term	Av SED	F pr
50 (109 μm)	$\Delta V1$	10.7	Site	4.2	0.424
	$\Delta V2$	10.8	ΔV	0.5	0.653
	$\Delta V3$	11.4	Site $\times$ ΔV	3.3	0.306
	$\Delta V4$	11.7			
65 (138 μm)	$\Delta V1$	10.7	Site	2.8	0.035
	$\Delta V2$	10.8	ΔV	0.5	0.172
	$\Delta V3$	11.4	Site $\times$ ΔV	2.4	0.051
	$\Delta V4$	11.7			

ΔV , volume increments; $\Delta V1$ is difference between V1 and V2; $\Delta V2$ is difference between V2 and V3; $\Delta V3$ is difference between V3 and V4; and $\Delta V4$ is difference between V4 and V5.

All volume increments are equal to 20% of the original core volume (see Fig. 1).

Av SED is average standard error of differences.

F prob, probability values for fixed terms are from Wald tests.

the lower resolution (109 μm) than those scanned at high resolution (30 μm). The maximum pore radii for cores scanned at 109 μm was 3500 and 2500 μm for the 0 to 10 and 20 to 30 soil depths respectively while for the 30 μm resolution scans maximum pore radii were 1200 and 700 μm for the respective depths.

3.2. Porosity characteristics measured using traditional methods and their comparison with the X-ray CT method

3.2.1. Effect of soil core diameter on porosity characteristics measured using a bulk density method

Mean values of soil bulk density, total porosity, field soil water content and air-filled porosity at the time of sampling were analysed for three soil core diameters (50 mm; 65 mm; 73 mm) sampled at two soil depths (0 to 10 cm; 20 to 30 cm) and across three sampling sites (Table 8). The effect of soil depth on these three properties was significant ($P < 0.001$). Soil bulk density in the 0 to 10 cm depth was significantly lower than that of the 20 to 30 cm depth, and hence corresponded to a higher porosity in the 0 to 10 cm depth. Soil water content was significantly greater in the 0 to 10 cm depth compared to the lower depth. Measures of bulk density and total porosity also differed depending upon core diameter with the 65 mm diameter cores having significantly lower bulk density than that measured in the 50 and 73 mm diameter cores. No significant differences were measured in soil water content for the three core diameters. Sampling site did not have an effect ($P > 0.05$) on bulk density or total porosity but the effect of site on soil–water content and air-filled porosity at the time of sampling was significant ($P < 0.001$). The effects of the interactions between site and core diameter on bulk density, total porosity and soil–water content were also significant ($P < 0.05$). Mean soil bulk density, total porosity, soil water content and air-filled porosity in the 0 to 10 and the 20 to 30 cm soil depths were 1.27 and 1.66 Mg m^{-3} , 52.0 and 37.5 vol.%, 32.0 and 14.7 vol.%, and 19.9 and 22.8 vol.%, respectively.

3.2.2. Porosity characteristics measured using a soil–water retention method and their comparison with the X-ray CT method

Mean porosity as a function of pore-diameter determined using soil–water retention method is presented in Table 9. The effect of soil depth on porosity was significant ($P < 0.033$) for all pore diameters except for pore diameters greater than 3 and 1 μm . Porosity in the 0 to 10 cm depth was significantly lower than that in the 20 to 30 cm depth. The effect of depth on porosity values for pore diameters greater than 3 and 1 μm was non-significant ($P > 0.05$). The effect of site on porosity was non-significant ($P > 0.05$).

Relationships between total porosity and macroporosity determined using either CT or soil–water retention methods were compared (Fig. 4). In the 0 to 10 cm soil depth, CT-measured macroporosity was greater

Table 7Porosity characteristics at three scanning positions (top, middle, bottom) and for the scans (30 μm) of the whole core of 50 mm soil cores collected at two soil depths.

Soil depth (cm)	Soil property	Scan position	Scan depth (mm)	Mean (LOG (soil property))	Mean	Fixed term	Av SEd	F pr
0–10	CT macroporosity ($\text{m}^3 \text{m}^{-3}$, %)	Top	14.5	2.2	8.8	Site	0.26	0.588
		Middle	31.5	2.3	10.4	Position	0.17	0.045
		Bottom	48.5	1.8	6.2	Site $\times$ position	0.35	0.122
		Whole core	31.5	2.2	8.7			
	Mean pore diameter (μm)	Top	14.5	4.46	86	Site	0.08	0.823
		Middle	31.5	4.47	87	Position	0.06	<0.001
		Bottom	48.5	4.36	78	Site $\times$ position	0.11	0.543
		Whole core	31.5	6.03	415			
	CT macroporosity ($\text{m}^3 \text{m}^{-3}$, %)	Top	214.5	2.05	7.8	Site	0.24	0.783
		Middle	231.5	1.93	6.9	Position	0.16	0.17
		Bottom	248.5	1.79	6.0	Site $\times$ position	0.32	0.939
		Whole core	231.5	1.72	5.6			
20–30	CT macroporosity ($\text{m}^3 \text{m}^{-3}$, %)	Top	214.5	4.50	90	Site	0.06	0.857
		Middle	231.5	4.49	89	Position	0.02	<0.001
		Bottom	248.5	4.46	86	Site $\times$ position	0.06	0.998
		Whole core	231.5	5.74	312			
	Mean pore diameter (μm)	Top	214.5	4.50	90	Site	0.06	0.857
		Middle	231.5	4.49	89	Position	0.02	<0.001
		Bottom	248.5	4.46	86	Site $\times$ position	0.06	0.998
		Whole core	231.5	5.74	312			
	CT macroporosity ($\text{m}^3 \text{m}^{-3}$, %)	Top	214.5	4.50	90	Site	0.06	0.857
		Middle	231.5	4.49	89	Position	0.02	<0.001
		Bottom	248.5	4.46	86	Site $\times$ position	0.06	0.998
		Whole core	231.5	5.74	312			

Av SEd is average standard error of differences.

F prob, probability values for fixed terms are from Wald tests.

than the macroporosity measured using the soil water retention, while in the 20 to 30 cm soil depth, CT-measured macroporosity values were lower. However, the CT-measured macroporosity followed a similar trend to that of the soil water retention method.

4. Discussion

4.1. Evaluation of within and between core sampling strategies for measuring porosity characteristics with X-ray CT

CT is a novel technique for investigating the porosity of compacted soils (Anderson et al., 2010; Borges et al., 2012; Cortina-Januchs et al., 2011; Kim et al., 2010; Munkholm et al., 2012, 2013; Vaz et al., 2011). However, the potential spatial resolution in a CT image is influenced by a number of interrelated factors including sample diameter (Ketchman and Carlson, 2001; Nielsen, 2004; Romans, 2012). While decreasing the sample diameter (i.e. diameter of the soil core) allows for increased spatial resolution, the smaller sample sizes increase the potential influence that collecting the soil core could have on soil structure, particularly around the edge of the core. This is especially problematic when trying to reconcile the often large sample sizes that are required to capture differences in the heterogeneous soil matrix, with the small spatial scale over which these differences may occur. We investigated how sampling strategies both within and between cores affect CT-based measures of porosity in loamy pasture soils. We hypothesised that a smaller diameter soil core may compact a larger percentage of the soil sample during the coring process hence leading to lower measures of porosity, in particular around the edges of the cores. We found, however, that the 65 mm diameter cores had only marginally greater CT measured macroporosity than that measured in the 50 mm diameter cores (Table 3). This trend was observed in the traditional bulk density based analyses (Table 8) where greater differences between the 65 and 50 mm cores were found. However the greater total porosity measured in the 65 mm diameter cores using the traditional method was not found in total porosity measured in 73 mm diameter cores, where the latter was not significantly different to that of the 50 mm diameter cores (Table 8). This discrepancy may reflect the bias that core size can have on the measurement of soil characteristics (Starr et al., 1995). However, the additional analysis of edge effect on CT macroporosity (Table 6) provides confidence in the soil coring technique as on a macro-scale, no significant compaction around the edges was observed in the 50 or 65 mm diameter soil cores. Further analysis of different volumes of images (Table 5) indicated that no compaction occurred in the area that was analysed in the 50 mm diameter cores when scanned at 109 μm .

Further comparisons were also used to investigate how the process of selecting a small volume in the centre of a soil core for analysis may affect measures of porosity. While such approaches are often used to reduce computationally demanding processes, it might bias results. Decreasing the volume of soil analysed in cores from 100 to 20% of the original volume did not affect measures of porosity. However, mean pore diameter decreased in 50 mm soil samples (Table 5). Higher

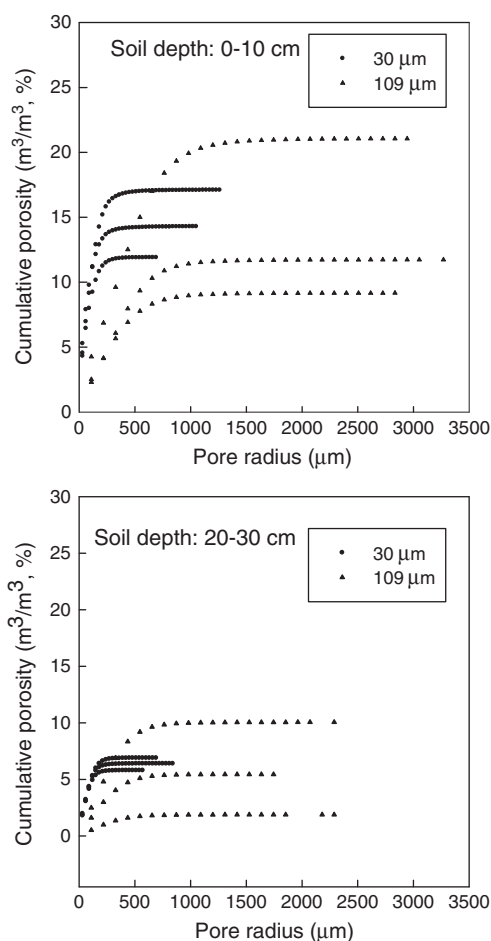


Fig. 3. Effect of scan resolution (30, 109 μm) on changes in porosity (m^3/m^3 , %) with pore radius (μm) measured for 50 mm soil cores at two soil depths (0 to 10, 20 to 30 cm) and three sampling sites (gateway, upper slope and lower slope). For 30 μm resolution, scans from the middle of cores were plotted.

Table 8

Mean soil bulk density, total porosity, soil–water content and air-filled porosity at the time of sampling of intact whole soil cores of three diameters (50, 65, 73 mm) collected at two soil depths (0 to 10, 20 to 30 cm) determined using a bulk density method.

Soil property	Core Diameter (mm)	Depth (cm)		Mean	Fixed term	Av SEd	F prob
		0–10	20–30				
Bulk density (mg m^{-3})	50	1.32	1.70	1.51	Site	0.02	0.871
	65	1.14	1.58	1.36	Depth	0.02	<0.001
	73	1.35	1.69	1.52	Core diameter	0.02	<0.001
	Mean	1.27	1.66		Site $\times$ Depth	0.04	0.068
					Site $\times$ Core diameter	0.04	0.002
Total porosity ($\text{m}^3 \text{m}^{-3}$, %)	50	50.0	35.8	42.9	Depth $\times$ Core diameter	0.04	0.164
	65	56.8	40.2	48.5	Site $\times$ depth $\times$ core diameter	0.06	0.039
	73	49.0	36.4	42.7	Site	0.8	0.871
	Mean	52.0	37.5		Depth	0.8	<0.001
					Core diameter	0.8	<0.001
Soil water content ($\text{m}^3 \text{m}^{-3}$, %)	50	31.5	14.2	22.9	Site $\times$ depth	1.3	0.068
	65	32.8	14.6	23.7	Site $\times$ core diameter	1.4	0.002
	73	31.8	15.2	23.5	Depth $\times$ core diameter	1.3	0.164
	Mean	32.0	14.7		Site $\times$ depth $\times$ core diameter	2.3	0.039
					Site	1.3	<0.001
Air-filled porosity ($\text{m}^3 \text{m}^{-3}$, %)	50	18.6	21.6	30.1	Depth	1.1	<0.001
	65	24.0	25.6	38.2	Core diameter	1.3	0.779
	73	17.2	21.2	37.1	Site $\times$ depth	1.9	0.013
	Mean	19.9	22.8		Site $\times$ core diameter	2.2	0.031
					Depth $\times$ core diameter	1.9	0.834
					Site $\times$ depth $\times$ core diameter	3.2	0.06
					Site	1.0	<0.001
					Depth	0.9	0.002
					Core diameter	1.0	<0.001
					Site $\times$ depth	1.5	0.168
					Site $\times$ core diameter	1.8	0.407
					Depth $\times$ core diameter	1.5	0.536
					Site $\times$ depth $\times$ core diameter	2.5	0.189

Air-filled porosity at the time of sampling was calculated as the difference between total porosity and soil water content.

Av SEd is average standard error of differences.

F prob, probability values for fixed terms are from Wald tests.

resolution (30 μm) scans of these cores also indicated that within soil cores, macroporosity and pore diameter were always lower at the bottom of the core (Table 7). These significant variations in measurements of macroporosity and pore diameter depending upon the sampling strategy used (both within and between cores) highlight the importance of carefully selecting sampling and scanning parameters that are relevant to the question being posed, and understanding the limitations (either physical or computational) of a given set of parameters. It must also be considered that CT scanning parameters and processing algorithms can also influence these measures, particularly when dealing with heterogeneous soil samples. Thus, while authors

have used core sizes of 150 mm diameter for characterising macroporosity of undisturbed soils using X-ray CT scans (e.g. Jassogne et al., 2007), their scan resolution for the whole core diameter was 300 μm which is considerably lower than the resolutions (109 and 138 μm) achieved in this study. Based on our above findings, we would recommend taking the whole-core scans of either 50 or 65 mm diameter cores for future measures of soil macroporosity using CT. A preference would be given to 50 mm diameter cores given the potential to measure porosity at a higher resolution. We would also recommend analysing the full volume of the scan, as while smaller sample volume did not affect measures of porosity, it biased measures of mean pore diameter and this could apply to other important pore parameters such as pore-space connectivity. While these parameters have been optimised for loamy pasture soils (and may hence differ depending upon factors including soil type, total soil porosity, heterogeneity and replication), the results hence provide a specific recommendation for future studies in similar soil types and/or for soil under similar uses. However, the sampling and scanning parameters that have been identified in this study may also provide a useful guide for other soil types and applications.

4.2. Effect of scanning resolution on porosity characteristics measured using X-ray CT

Many of the CT-measured macroporosity results have been reported for a whole core length (e.g. Anderson et al., 2010) and very few studies reported variations in macroporosity within collected samples (Asare et al., 2001; Kim et al., 2010; Nielsen, 2004). In this study, the effects of sample volume and resolution within a soil core were also examined. It is generally expected that cores scanned at higher resolutions should have more macroporosity than those cores scanned at lower resolutions. Interestingly, when small areas (10.5 $\times$ 10.5 $\times$ 15 mm) within the 50 mm cores were scanned at the highest resolution possible for that core diameter (30 μm); macroporosity of the whole core was

Table 9

Pore size distribution of intact whole core samples at two soil depths determined using a soil–water retention method.

Matric potential (cm)	Pore diameter (μm) ^a	Mean porosity ($\text{m}^3 \text{m}^{-3}$, %) at two soil depths (cm) ^b		F pr
		0–10	20–30	
10	>298	3.9	6.3	0.033
20	>149	4.3	6.5	0.007
50	>60	5.7	8.1	0.017
100	>30	7.4	9.9	0.018
300	>10	10.1	12.6	0.017
600	>5	11.4	13.6	0.021
1000	>3	12.8	15.2	0.057
3000	>1	15.2	17.7	0.115
15,000	>0.2	42.7	31.3	<0.001

F prob, probability values for fixed terms are from Wald tests.

^a Pore diameter (d) was calculated as $h = 0.298/d$ where h is the matric potential (Eq. (1)).

^b Porosity was calculated as the difference between total porosity and soil–water content at a given matric potential.

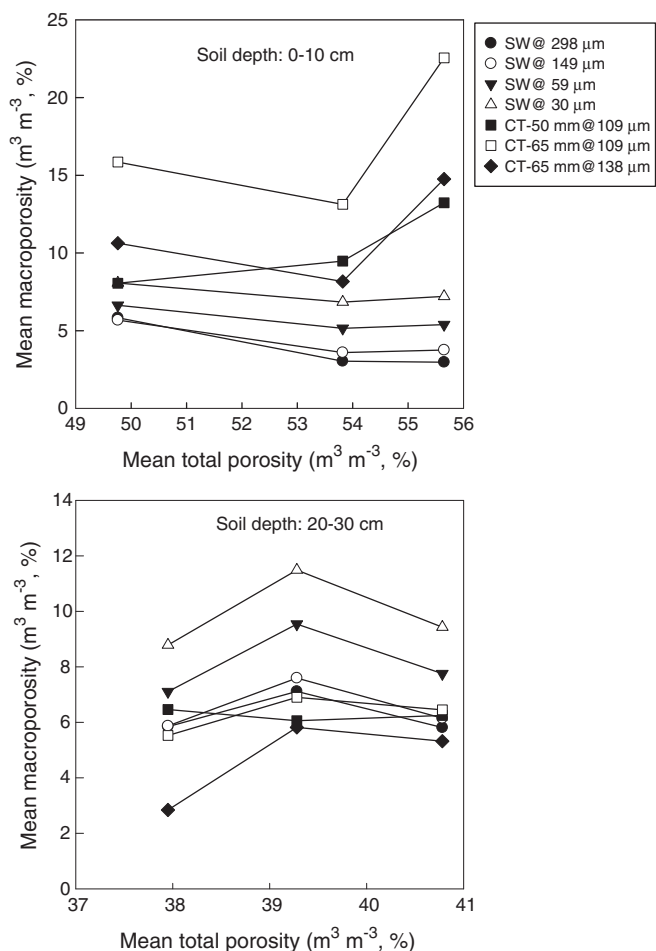


Fig. 4. Mean values of macroporosity as a function of mean total porosity at three sites using the soil water retention (SW) at (30, 59, 149, 298 μm) and the X-ray CT methods scanned at 109 and 138 μm for two soil depths (0 to 10, 20 to 30 cm).

only found to be marginally different (0–10 cm) or not different (20–30 cm) to that measured in larger samples (whole cores) scanned at 109 μm. These findings suggest that the amount of the pore space between 30 and 109 μm radius may have contributed less to soil porosity but such a result would differ depending upon the pore size distribution of a given soil. The manual adjustment of the threshold may be different for large and small pores and may have also contributed to this result. However, two counteracting factors need to be considered when using CT scans. While smaller pores can be detected at higher resolution (i.e. 30 μm cf 109 μm) in smaller cores, these smaller sample sizes may not reflect the heterogeneity evident in soils.

The difference in maximum pore radius for cores scanned at 109 μm compared with that for the 30 μm resolution scanned cores (Fig. 3) could be because a much smaller area (10.5 × 10.5 × 15 mm) was scanned compared to whole core area (Table 2) and therefore, the area sampled (because it is much smaller) may not have captured the full range of pore diameters or only part of large pores that fell within the sample. Fig. 3 clearly shows that the cumulative values asymptote early on, indicating that no larger pores are detected before the cut-off, whereas they are present in the larger sample.

The macroporosity values at various pore-size classes (218 to >2000 μm pore diameters) were also considered for evaluating the CT method for measuring porosity characteristics (Table 4). With maximum resolutions of 109 and 138 μm achievable for whole-core scans of 50 and 65 mm diameter cores respectively, we categorised pores for these two core sizes based on a common pore diameter of

300 μm. We hypothesised that macroporosity values greater than 300 μm diameter pores would be the same for cores scanned at both 109 and 138 μm. However, macroporosity values >300 μm diameter pores were found to be significantly different between these two resolutions. These differences might be due to differences in core size diameters, sample to sample variation or because different resolutions have different powers to classify pore-space within a given pore class.

While in this study we found that absolute measures of macroporosity might not change with core size, or the volume of soil analysed, the characteristics of the pore-space that is captured differs significantly (as demonstrated by changes in mean pore diameter). Hence these results highlight the need for targeted approaches when using CT to quantify pore-space characteristics. Often the resolution used is determined by the resolution of the CT machine available, or the diameter of the soil cores. However, there is a need to understand the scale at which the treatment differences are expected to occur, and hence the volume of sample, scan resolution and number of replicate cores required to capture these differences. It must also be acknowledged that any reported results are a product of these parameters. Further research is required for different soil types and land uses where the effect of sample volumes and scanning resolutions on CT macroporosity distribution are visualised and quantified by scanning the sample area within the sample at various resolutions, and mapping this into the same area when it is scanned at the lower resolution.

4.3. Comparison of X-ray CT with a traditional method for measuring porosity in dairy pasture soils and implications for compaction

In comparing CT-measured porosity characteristics with a traditional method for determining porosity using bulk density, the results for both methods were comparable. No significant differences in both porosity and pore diameter were found between the three sites. Given that no differences were detected between sites with either the X-ray CT or traditional analytical method, it is unlikely that limitations in either technique are preventing detection of site differences. The sites were selected in three distinct locations in the field to represent moderate (upper- and lower slope locations) and high trafficking (near gateway). We expected that macroporosity would be lower near the gateway due to greater frequency of cow movement as the herd enters and leaves at least twice daily when the paddock is grazed. We deliberately avoided bare and heavily compacted areas on the gateway where no obvious pasture growth was observed, and consequently may have reduced the potential to detect differences in porosity between sites. The number of replication for sites may also have been inadequate to capture differences in porosity. Alternatively, the soils may have already been compacted to the same extent during the five years of continuous grazing since pasture renovation in 2007. This is consistent with findings by various authors that most compaction usually occurs during the initial periods of trampling, in particular, immediately after renovation when soil is relatively less dense, because as the soil becomes compacted its bearing strength increases making it less susceptible to further compaction (Bell et al., 2011; Greenwood and McKenzie, 2001).

Macropores are defined by many authors (e.g. Drewry et al., 2008) as pores having diameters larger than 30 μm. Traditionally, macroporosity is estimated as the difference between total porosity and water-filled porosity at an equivalent pore diameter of 30 μm (Drewry et al., 2008). In this study, for the CT-method, the values of porosity were determined for an equivalent pore diameter of 60 (30 μm resolution), 218 (109 μm resolution) or 276 (138 μm resolution) μm or greater using X-ray CT. Therefore, according to the above definition any pores detected using the CT scan at these resolutions are considered macropores. Since the CT measured macroporosity accounted for all pores having diameter larger than 60 μm, while traditional methods account all pores larger than 30 μm, it was hypothesised that the values of CT macroporosity may be lower compared to traditionally determined macroporosity for a given land-use, soil type and climatic conditions. In contrast, we

found the values of CT macroporosity to be higher than those from the soil–water retention method especially, in the 0 to 10 cm soil depth. Water-filled porosity at an equivalent pore diameter, say for example, of 30 μm is determined by draining water from a saturated intact core sample using a matric suction of 100 cm. Water is drained through connected soil pores and therefore, unconnected pores are not accounted for in the soil water retention method, while the latter would be measured using the X-ray CT. The connected and unconnected pores could easily be visualised and quantified using the X-ray CT method (e.g. Mooney, 2002; Munkholm et al., 2012; Tracy et al., 2012) while this is not possible using soil–water retention method.

In some situations, soil compaction may close or reduce the macroporosity and pore connectivity and the traditional method may not be sensitive enough to detect the small reduction in bulk density. In those situations, total porosity as determined from the soil bulk density method would be lower compared to its true value. Since the macroporosity is determined as the difference between total porosity and soil–water content at a specified matric potential (given pore diameter) from soil–water retention curves, the macroporosity determined using this method is likely to be lower than the true value. On the other hand, CT macroporosity values are independent of total porosity values.

The X-ray CT method also has benefits in the additional spatial information that it can provide. Spatial distribution of pores, distribution of pores greater than 300 μm diameter pores and pore–space connectivity may also be calculated to provide a better understanding of soil structure and how the size and arrangement of the pores may influence soil processes. In this study the X-ray CT analysis demonstrated that mean pore diameter decreased with soil depth. Since macroporosity is traditionally determined for a whole sample length of about 63–76 mm (Anderson et al., 2010; Rab et al., 2009), the micro-variation of macroporosity characteristics within the collected sample cannot be studied in detail. Another advantage of CT is that this method can be used to determine vertical variations within collected cores which is very important for understanding the soil–root interface. Therefore, the CT-method is likely to be a better method than the soil–water retention method for determining soil macroporosity, in particular porosity larger than 300 μm diameter.

Soil compaction may reduce macroporosity to the extent that growth or even survival of roots is limited by oxygen availability, in particular when the soil is wet (Drewry et al., 2008; Greacen and Sands, 1980; Lipiec and Hatano, 2003). This will occur when the oxygen requirements for respiration in the soil exceed the rate at which oxygen in the soil air can be replaced from the atmosphere. Some authors proposed threshold values of macroporosity of 10–15%, below which, soil aeration is likely to be reduced (Cannell, 1977; Carter, 1988; Grable, 1971; Gradwell, 1965; Greenwood, 1975; Hodgson and MacLeod, 1989; Lipiec and Hatano, 2003). Macroporosity values found in this study for both soil depths determined using the CT and soil–water retention methods respectively were 11.2 and 5.2 vol.% (Table 3), and 7.4 and 9.9 vol.% (Table 9) which suggest that root growth in these loamy pasture soils may be moderately to severely affected due to low oxygen availability in the 0–10 and 20–30 cm soil depths respectively under wet conditions such as may occur after heavy rainfall.

5. Conclusions

The X-ray CT method presents a valuable tool for providing rapid, high-throughput quantification and characterisation of porosity in soils, in particular for pores larger than 300 μm diameter. The technique benefits over traditional bulk density based methods on the ability to characterise and quantify not only porosity, but also the diameter, connectivity and the spatial distribution of this pore-space on a micro-scale. These are important parameters for understanding the implications of pore size and their continuity on soil water dynamics and subsequent impacts on plant productivity and the environment. In comparing

a range of within (sample volume, resolution) and between core (core diameter) sampling strategies for characterising porosity of loamy pasture soils using X-ray CT scans of intact cores, we identified that macroporosity, mean pore diameter and pore size distribution varied significantly depending upon the sampling and CT parameters that were selected. The results highlight the importance of having targeted sampling and analytical parameters based on specific research aims. The results give confidence in using a range of core diameters (50 or 65 mm) for measuring macroporosity but highlight the importance of maximising the volume of soil that can be analysed within the core, to better estimate parameters such as mean pore diameter. The results also demonstrate the significant influence that sampling strategies can have on CT-based measures of porosity. Nevertheless, it is concluded that X-ray CT is a valuable tool for characterising pore size distribution and pore diameter from the macro- (between cores) to the micro-scale (within a soil core). Based on our comparison of CT macroporosity with soil water retention (SWR) method we conclude that the CT-method is likely to be better than the SWR method for determining soil macroporosity larger than 300 μm diameter and their continuity of loamy pasture soils. Further research is required for different soil types and land uses where the effect of sample volumes and scanning resolutions on CT macroporosity distribution are visualised and quantified by scanning the sample area within the sample at various resolutions, and mapping this into the same area when it is scanned at the lower resolution.

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